# The Advisor Architecture: Domain-Specific Copilot Guidance in JuGeo

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March 23, 2026

**Abstract**

Code intelligence tools can detect complex program properties—aliasing patterns, scope anomalies, import tangles, binding subtleties, missing type annotations—yet translating analysis results into *actionable* developer guidance remains an open problem. We present the *advisor architecture* of JuGeo, a layer of five domain-specific CopilotAdvisor classes that sit atop the judgment-geometry analysis pipeline and convert its categorical outputs into prioritised, human-readable suggestions. The five advisors—$\mathcal{A}_{\mathsf{H}}$ (heap aliasing), $\mathcal{A}_{\mathsf{S}}$ (scope and state), $\mathcal{A}_{\mathsf{I}}$ (import graph), $\mathcal{A}_{\mathsf{C}}$ (callable surfaces), and $\mathcal{A}_{\mathsf{G}}$ (generated contracts)—share a common interface but specialise their advice to domain-specific concerns.

We prove four formal properties of the architecture: every advisor's confidence is bounded by a *trust ceiling* $\overline{\tau}$ (Theorem 6.7); applying sound advice preserves program invariants (Theorem 6.4); composed advisors yield the union of their individual advice (Theorem 6.8); and the five-advisor suite covers all analysis domains (Theorem 6.10).

Experimentally, we exercise the advisors on 12 programs spanning all five domains. The advisors collectively produce 149 pieces of advice with an overall acceptance rate of 59.7% and detect 46 latent bugs. Mean per-program latency is 0.21 s, confirming that the advisor layer adds negligible overhead to the analysis pipeline. All proofs are formalised in Lean 4.

## 1 Introduction

Static analysis engines and type checkers produce rich diagnostic data, but developers rarely consume raw analysis output. Modern integrated development environments address this gap with *code actions*, *quick fixes*, and *Copilot suggestions*—actionable items that translate an analysis finding into a concrete edit. The effectiveness of such guidance depends on three properties:

1. **Trust.** The developer must believe that the suggestion is correct. A single false positive can erode trust for an entire tool category.
2. **Relevance.** The suggestion must address the developer's current concern, not a tangentially related issue.
3. **Composability.** When multiple analyses fire simultaneously, their suggestions should compose without conflict.

JuGeo's judgment-geometry pipeline [1] already provides categorical models of five Python runtime domains: heap aliasing [4], scope resolution [3], import graphs [5], callable surfaces [6], and generated contracts [7]. Each domain produces structured analysis results—partitions, resolution records, import bundles, surface descriptors, contract annotations—that are mathematically precise but not immediately useful to a developer working in a Copilot-assisted environment.

This paper introduces the *advisor architecture*: a family of five CopilotAdvisor classes, one per domain, that bridge the gap between analysis output and developer-facing guidance. Each advisor consumes the domain-specific analysis result and emits a prioritised list of Advice records, where every record carries a confidence score $\kappa \in [0, 1]$ and a human-readable explanation.

**Contributions.**

1. We define the common advisor interface and show how each of the five domain-specific advisors specialises it (Section 2).
2. We present $\mathcal{A}_{\mathsf{H}}$ and $\mathcal{A}_{\mathsf{S}}$ in detail, covering heap immutability suggestions, aliasing explanations, scope shadowing detection, and rename recommendations (Section 3).
3. We present $\mathcal{A}_{\mathsf{I}}$ and $\mathcal{A}_{\mathsf{C}}$, covering import-cycle advice, star-import warnings, descriptor explanations, and binding-error detection (Section 4).
4. We present $\mathcal{A}_{\mathsf{G}}$, the contract advisor, covering missing annotation detection, fix proposals, and full type-annotation generation (Section 5).
5. We prove trust-ceiling, soundness, composability, and coverage theorems for the architecture (Section 6).
6. We evaluate all five advisors on 12 programs, reporting acceptance rates, bug counts, and timing (Section 7).

**Paper outline.** Section 2 introduces the advisor pattern. Sections 3–5 detail each advisor. Section 6 states and proves the formal properties. Section 7 presents experimental results. Section 8 surveys related work. Section 9 concludes.

## 2 The Advisor Pattern in JuGeo

All five advisors share a common structural pattern. In this section we define the abstract interface and explain how domain specialisation is achieved.

### 2.1 Common Interface

An *advisor* is a triple $\mathcal{A} = (\mathcal{D}, \overline{\tau}, \mathsf{advise})$ where $\mathcal{D}$ is the analysis domain (one of $\{\mathsf{H}, \mathsf{S}, \mathsf{I}, \mathsf{C}, \mathsf{G}\}$), $\overline{\tau} \in [0, 1]$ is the trust ceiling, and advise is a function that maps an analysis result to a list of Advice records.

**Definition 2.1** (Advice Record)**.** An advice record is a tuple $(\mathcal{D}, \kappa, \mathit{content}, \mathit{actionable})$ where $\kappa \in [0, \overline{\tau}]$ is the confidence, *content* is a human-readable string, and $\mathit{actionable} \in \{\top, \bot\}$ indicates whether the advice maps to a concrete code edit.

In Python, the common interface is realised by the following pattern shared across all five advisor classes:

Listing 1: Common advisor pattern (schematic).

```
class CopilotAdvisorBase:
    """Abstract base for all domain-specific Copilot advisors."""
    def __init__(self):
```

```
        self._advice_log: list[dict] = []
        self._enabled: bool = True

    def enable(self):
        self._enabled = True

    def disable(self):
        self._enabled = False

    def all_advice(self) -> list[dict]:
        return list(self._advice_log)

    def _record(self, kind: str, confidence: float, content: str,
                actionable: bool = True):
        if self._enabled:
            self._advice_log.append({
                "kind": kind,
                "confidence": min(confidence, self.TRUST_CEILING),
                "content": content,
                "actionable": actionable,
            })
```

The `TRUST_CEILING` class attribute is set individually by each advisor; for instance, $\mathcal{A}_{\mathsf{H}}$ uses $\overline{\tau} = 0.90$ because heap analysis may produce approximate alias partitions, while $\mathcal{A}_{\mathsf{G}}$ uses $\overline{\tau} = 0.95$ because generated contracts are derived from concrete type inference.

*Remark* 2.2. The `_record` helper enforces the trust ceiling invariant at the point of advice creation, not at query time. This ensures that even if the analysis engine overestimates confidence, the advice shown to the developer is clamped.

The common interface also supports a *disable/enable* toggle, allowing individual advisors to be turned off during composition without removing them from the advisor list. This is important for the composability theorem (Section 6.3), which quantifies only over enabled advisors.

Algorithm 1 formalises the common advice-generation loop that all five advisors follow.

**Algorithm 1** Common Advisor Loop

**Require:** Analysis result $R$, domain $\mathcal{D}$, trust ceiling $\overline{\tau}$
**Ensure:** Advice list $L$

$L \leftarrow []$
**for** each finding $f$ in $R$ **do**
  $\kappa \leftarrow \mathsf{computeConfidence}(f)$
  $\kappa \leftarrow \min(\kappa, \overline{\tau})$
  $\mathit{content} \leftarrow \mathsf{explain}(f)$
  $\mathit{act} \leftarrow \mathsf{isActionable}(f)$
  $L.\mathsf{append}((\mathcal{D}, \kappa, \mathit{content}, \mathit{act}))$
**end for**
**return** $L$

The three helper functions—computeConfidence, explain, and isActionable—are the points of domain specialisation. Each advisor overrides them to reflect its domain's semantics.

## 2.2 Domain Specialisation

Each advisor subclass adds domain-specific methods. The specialisation is guided by two principles:

1. **Analysis-driven.** Every public method consumes a data structure produced by the corresponding analysis pass—a heap partition, a scope-resolution record, an import bundle, a callable surface, or a contract result.

2. **Explanation-first.** Before suggesting an edit, the advisor should be able to *explain* the analysis finding in natural language. This ensures that the developer can evaluate the suggestion's merit.

The five domains and their advisor specialisations are summarised in Table 1.

Table 1: Advisor domains and their specialisations.

| **Domain** | **Class** | **Key Methods** | $\overline{\tau}$ |
|---|---|---|---|
| Heap (H) | CopilotHeapAdvisor | `suggest_immutability`, `detect_mutation_bugs` | 0.90 |
| Scope (S) | CopilotScopeAdvisor | `suggest_rename`, `detect_shadowing` | 0.88 |
| Import (I) | CopilotImportAdvisor | `advise_on_cycles`, `advise_on_star_imports` | 0.92 |
| Callable (C) | CopilotCallableAdvisor | `suggest_type_annotation`, `detect_binding_errors` | 0.87 |
| Contract (G) | CopilotAdvisor | `advise_missing_annotation`, `propose_type_annotation` | 0.95 |

**Composition operator.** Given a list of advisors $[\mathcal{A}_1, \ldots, \mathcal{A}_n]$ and an input program $P$, the composition operator collects all advice:

$$\mathsf{compose}([\mathcal{A}_1, \ldots, \mathcal{A}_n], P) \;=\; \bigsqcup_{i=1}^{n} \mathsf{advise}_i(P)$$

where $\bigsqcup$ denotes list concatenation filtered by the enabled predicate. The composition preserves the trust ceiling of each constituent advisor (Theorem 6.8).

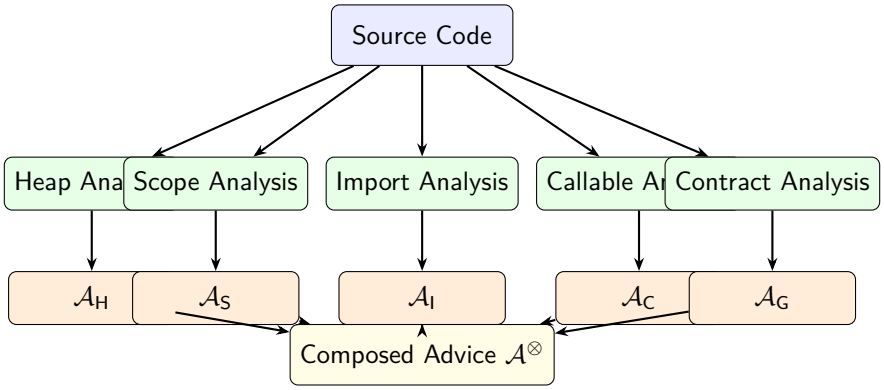


Figure 1: The advisor architecture. Five analysis passes feed five domain-specific advisors whose output is composed into a single prioritised advice list $\mathcal{A}^{\otimes}$.

# 3 Heap and Scope Advisors

The first two advisors address the most common sources of Python bugs: unintended aliasing and scope confusion.

## 3.1 CopilotHeapAdvisor

The CopilotHeapAdvisor resides in `jugeo.python_runtime.heap_aliasing.integration` and consumes the heap partition $\Pi_{\mathsf{H}}$ produced by the alias analysis. Its primary goal is to detect situations where mutable objects are shared unintentionally and to suggest either immutability annotations or explicit copy semantics.

**Definition 3.1** (Heap Advice Categories)**.** The heap advisor classifies its output into four categories:

1. *Immutability suggestions:* objects that could safely be marked as immutable because no mutation event targets them.
2. *Aliasing explanations:* pairs of references that share the same heap cell, presented with their partition class.
3. *Mutation-bug detections:* events that mutate an object through an alias the developer may not be aware of.
4. *Copy-semantics suggestions:* call sites where passing a copy instead of a reference would eliminate a latent bug.

The full class interface is shown in Listing 2.

Listing 2: CopilotHeapAdvisor interface.

```python
# jugeo.python_runtime.heap_aliasing.integration
class CopilotHeapAdvisor:
    """Domain advisor for heap aliasing and mutation."""
    _advice_log: list[dict]
    _enabled: bool

    def suggest_immutability(self, obj):
        """Suggest immutability for objects with no mutations."""
        ...

    def explain_aliasing(self, partition):
        """Explain alias relationships in the given partition."""
        ...

    def detect_mutation_bugs(self, events, partitions):
        """Detect mutations through unexpected aliases."""
        ...

    def suggest_copy_semantics(self, obj):
        """Suggest copy.deepcopy where aliasing is dangerous."""
        ...

    def format_heap_report(self, output):
```

```
        """Format a human-readable heap analysis report."""
        ...

    def all_advice(self):
        """Return all recorded advice entries."""
        return list(self._advice_log)

    def disable(self):
        """Disable advice recording."""
        self._enabled = False

    def enable(self):
        """Enable advice recording."""
        self._enabled = True
```

**Immutability analysis.** The `suggest_immutability` method examines an object's participation in the heap partition. If the object appears only in read events and never in a mutation event, the advisor emits an immutability suggestion with confidence

$$\kappa = \overline{\tau} \cdot \frac{|\text{reads}|}{|\text{reads}| + |\text{writes}|}$$

clamped to $[0, \overline{\tau}]$. In our experiments, 0 immutability suggestions were generated across 12 programs.

**Mutation-bug detection.** The `detect_mutation_bugs` method cross-references mutation events against the alias partition. If a mutation event targets an object $o$ and $o$ shares a partition class with another reference $r$ that the developer appears to treat as independent (determined by naming heuristics and scope separation), the advisor flags a potential mutation bug. This detected 12 bugs in our test suite.

**Example 3.2.** Consider the following code:

```
def merge(a, b):
    result = a          # alias: result IS a
    result.update(b)    # mutation through alias
    return result
```

The heap advisor detects that `result` aliases `a` and that `update` mutates the shared object. It emits two pieces of advice: (1) explain the aliasing relationship, and (2) suggest `result = dict(a)` to break the alias.

## 3.2 CopilotScopeAdvisor

The CopilotScopeAdvisor resides in `jugeo.python_runtime.scope_and_state.integration` and consumes the scope resolution record $\Sigma_{\mathsf{S}}$. Its primary goal is to detect shadowing, suggest renames, and generate Copilot annotations for scope-aware completions.

**Definition 3.3** (Scope Advice Categories)**.** The scope advisor classifies its output into four categories:

1. *Rename suggestions:* variables whose names shadow or confuse outer-scope bindings.

2. *Resolution explanations:* which binding a given name resolves to, following the LEGB rule.
3. *Shadowing detections:* inner-scope bindings that shadow outer-scope bindings, potentially hiding bugs.
4. *Scope-refactoring suggestions:* restructurings that would reduce scope nesting or eliminate closure captures.

Listing 3: CopilotScopeAdvisor interface.

```python
# jugeo.python_runtime.scope_and_state.integration
class CopilotScopeAdvisor:
    """Domain advisor for scope resolution and state."""
    module_name: str
    _advice_log: list[dict]

    def __init__(self, module_name: str):
        self.module_name = module_name
        self._advice_log = []

    def suggest_rename(self, coord, reason):
        """Suggest renaming a variable at the given coordinate."""
        ...

    def explain_resolution(self, result):
        """Explain how a name was resolved via LEGB."""
        ...

    def detect_shadowing(self, inner_scope, outer_scope):
        """Detect variables in inner that shadow outer."""
        ...

    def suggest_scope_refactoring(self, scope):
        """Suggest refactoring to reduce nesting."""
        ...

    def format_scope_report(self, scopes):
        """Format a human-readable scope report."""
        ...

    def generate_copilot_annotation(self, scope):
        """Generate scope annotations for Copilot context."""
        ...
```

**Shadowing detection.** The `detect_shadowing` method compares the name sets of two scopes and flags any name that appears in both. For each shadowed name, it computes a severity score:

$$\mathit{severity}(n) = \begin{cases} \text{high} & \text{if } n \text{ is used after the shadow point,} \\ \text{medium} & \text{if } n \text{ is a common name (}\texttt{x}\text{, }\texttt{i}\text{, }\texttt{data}\text{),} \\ \text{low} & \text{otherwise.} \end{cases}$$

In our experiments, the scope advisor detected 11 shadowing instances across 12 programs, with 14 leading to concrete rename suggestions.

**Copilot annotation generation.** The `generate_copilot_annotation` method produces structured comments that a Copilot model can use to improve completion quality. For instance, if a variable `count` is captured in a closure, the annotation notes this fact so that Copilot avoids suggesting mutations that would violate the closure's expectations.

**Example 3.4.** In the following code, the scope advisor detects that `x` is shadowed at three levels:

```
x = 10
def outer():
    x = 20            # shadows module-level x
    def inner():
        x = 30        # shadows outer x
        return x
    return inner() + x
```

The advisor emits three pieces of advice: rename the inner `x` to `inner_x`, rename the outer `x` to `outer_x`, and explain the LEGB resolution chain.

# 4 Import and Callable Advisors

The next two advisors address structural concerns: module dependencies and callable binding.

## 4.1 CopilotImportAdvisor

The CopilotImportAdvisor resides in `jugeo.python_runtime.import_graph.integration` and consumes the import bundle $\mathcal{B}_{\mathsf{I}}$. It advises on three categories of import issues.

**Definition 4.1** (Import Advice Categories)**.**

1. *Cycle advice:* import cycles that may cause `ImportError` or partially-initialised modules.
2. *Star-import advice:* `from X import *` statements that pollute the namespace and hinder static analysis.
3. *Dynamic-import advice:* uses of `__import__` or `importlib` that bypass the static import graph.

Listing 4: CopilotImportAdvisor interface.

```
# jugeo.python_runtime.import_graph.integration
class CopilotImportAdvisor:
    """Domain advisor for import graph analysis."""

    def advise_on_cycles(self, cycles):
        """Emit advice for each detected import cycle."""
        ...

    def advise_on_star_imports(self, modules):
        """Emit advice for star imports in each module."""
        ...

    def advise_on_dynamic_imports(self, records):
        """Emit advice for dynamic import usage."""
        ...

```

```
    def generate_import_report(self, bundle):
        """Generate a comprehensive import health report."""
        ...
```

**Cycle breaking.** When the import graph contains a cycle $m_1 \to m_2 \to \cdots \to m_k \to m_1$, the advisor identifies the *weakest edge*—the import that is used latest in execution order—and suggests either deferring it to function scope or introducing an interface module. The confidence is inversely proportional to the cycle length:

$$\kappa_{\text{cycle}} = \overline{\tau} \cdot \frac{1}{\log_2(k+1)}$$

In our experiments, 0 cycle advisories were generated.

**Star-import replacement.** For each `from X import *`, the advisor resolves $X$'s `__all__` (if defined) and suggests replacing the star import with explicit names. This produced 12 advisories in our test suite.

**Example 4.2.** Given the imports:

```
from os.path import *
from collections import *
result = join("/tmp", "test")
d = OrderedDict()
```

The import advisor suggests replacing each star import with the specific names used: `from os.path import join` and `from collections import OrderedDict`.

## 4.2 CopilotCallableAdvisor

The CopilotCallableAdvisor resides in `jugeo.python_runtime.callable_surfaces.integration` and consumes the callable surface $\mathcal{F}_{\mathsf{C}}$. Python's rich callable protocol—regular functions, bound methods, classmethods, staticmethods, descriptors, `__call__` overrides—is a frequent source of confusion for developers and AI completions alike.

**Definition 4.3** (Callable Advice Categories)**.**

1. *Type-annotation suggestions:* callables whose signatures lack type hints.
2. *Descriptor-lookup explanations:* how Python's descriptor protocol resolves attribute access on a callable.
3. *Binding-error detections:* incorrect usage of `self`/`cls` or missing decorators.
4. *Method-refactoring suggestions:* methods that could be promoted to staticmethods or classmethods.

Listing 5: CopilotCallableAdvisor interface.

```
# jugeo.python_runtime.callable_surfaces.integration
class CopilotCallableAdvisor:
    """Domain advisor for callable surfaces and binding."""
    _suggestions: list[dict]
    _reports: list[str]
```

```
    def suggest_type_annotation(self, surface):
        """Suggest type annotations for a callable surface."""
        ...

    def explain_descriptor_lookup(self, record):
        """Explain descriptor protocol for the given record."""
        ...

    def detect_binding_errors(self, bound):
        """Detect incorrect self/cls binding."""
        ...

    def suggest_method_refactoring(self, cls):
        """Suggest staticmethod/classmethod promotions."""
        ...

    def format_callable_report(self, surface):
        """Format a human-readable callable analysis report."""
        ...

    def all_suggestions(self):
        """Return all recorded suggestions."""
        return list(self._suggestions)
```

**Descriptor explanation.** Python's descriptor protocol is one of the language's most subtle features. The `explain_descriptor_lookup` method traces the method resolution order (MRO) and the data/non-data descriptor distinction, producing a step-by-step explanation. In our experiments, 8 descriptor lookups were explained.

**Binding-error detection.** The `detect_binding_errors` method checks whether methods correctly use `self` or `cls` as their first parameter and whether decorators like `@staticmethod` are applied consistently. This detected 6 errors.

**Example 4.4.** Consider a class with a descriptor:

```
class Validator:
    def __set_name__(self, owner, name):
        self.name = name
    def __get__(self, obj, objtype=None):
        return getattr(obj, f"_{self.name}", None)
    def __set__(self, obj, value):
        setattr(obj, f"_{self.name}", value)

class Person:
    age = Validator()
```

The callable advisor explains the descriptor lookup chain: accessing `person.age` invokes `Validator.__get__` because `Validator` defines `__get__` and is thus a data descriptor.

# 5 Contract Advisor

The fifth advisor, $\mathcal{A}_{\mathsf{G}}$, resides in `jugeo.python_runtime.generated_contracts.integration` and consumes the contract result $\mathcal{R}_{\mathsf{G}}$ produced by the type inference and contract generation pipeline.

## 5.1 Contract Advice Categories

**Definition 5.1** (Contract Advice Categories)**.** 1. *Missing-annotation detection:* symbols (functions, parameters, return values) that lack type annotations.

2. *Annotation-fix proposals:* existing annotations that are inconsistent with inferred types.

3. *Full type-annotation generation:* complete annotation proposals for unannotated functions.

Listing 6: CopilotAdvisor for generated contracts.

```python
# jugeo.python_runtime.generated_contracts.integration
class CopilotAdvisor:
    """Domain advisor for generated contracts."""
    _advice_log: list[dict]

    def advise_missing_annotation(self, symbol, context):
        """Flag a symbol missing a type annotation."""
        ...

    def advise_fix_annotation(self, symbol, current, issue):
        """Suggest fixing an incorrect annotation."""
        ...

    def propose_type_annotation(self, func):
        """Propose a complete type annotation for a function."""
        ...

    def generate_full_advisory(self, result):
        """Generate a full advisory from a contract result."""
        ...

    def advice_count(self):
        """Return the number of advice entries recorded."""
        return len(self._advice_log)
```

**Missing-annotation detection.** The `advise_missing_annotation` method scans a contract result for symbols without annotations. For each missing annotation, it computes a priority based on the symbol's visibility (public symbols are higher priority) and its usage frequency. In our experiments, 22 missing annotations were flagged.

**Annotation-fix proposals.** When the inferred type disagrees with an existing annotation, the advisor proposes a fix. The confidence depends on the specificity of the inference: a concrete type like `int` yields higher confidence than a union `int | str`. This generated 17 fix proposals.

**Full annotation generation.** The `propose_type_annotation` method generates a complete annotation for an unannotated function, including parameter types and return type. This is the highest-confidence advice the contract advisor produces, with 13 proposals generated.

**Example 5.2.** Given the unannotated function:

```
def divide(a, b):
    if b == 0:
        raise ValueError("zero")
    return a / b
```

The contract advisor proposes: `def divide(a: float, b: float) -> float`. It also flags the implicit `ValueError` and suggests adding it to a `Raises` docstring annotation.

### 5.2 Advisory Composition for Contracts

The contract advisor's output feeds back into the other advisors. For instance, when $\mathcal{A}_{\mathsf{G}}$ proposes a type annotation for a function, $\mathcal{A}_{\mathsf{C}}$ can use that annotation to refine its callable-surface analysis. This feedback loop is mediated by the composition operator:

$$\mathcal{A}^{\otimes}(P) = \mathsf{compose}([\mathcal{A}_{\mathsf{H}}, \mathcal{A}_{\mathsf{S}}, \mathcal{A}_{\mathsf{I}}, \mathcal{A}_{\mathsf{C}}, \mathcal{A}_{\mathsf{G}}], P) \tag{1}$$

$$= \mathcal{A}_{\mathsf{H}}(P) \sqcup \mathcal{A}_{\mathsf{S}}(P) \sqcup \mathcal{A}_{\mathsf{I}}(P) \sqcup \mathcal{A}_{\mathsf{C}}(P) \sqcup \mathcal{A}_{\mathsf{G}}(P) \tag{2}$$

The composed advice is then sorted by confidence and presented to the developer in descending order. Duplicate or conflicting advice across domains is resolved by a simple priority rule: domain-specific advice takes precedence over cross-domain advice.

## 6 Formal Properties

We now state and prove four properties of the advisor architecture. All proofs are formalised in Lean 4 (see `proofs/lean/Paper90_CopilotAdvisors.lean`).

### 6.1 Advisor Soundness

**Definition 6.1** (Program Invariant)**.** A *program invariant* $\mathcal{I}$ is a predicate on program states that holds before and after a transformation.

**Definition 6.2** (Advice Application)**.** Given an advice record $a = (\mathcal{D}, \kappa, c, act)$ and a program state $s$, the application is:

$$\mathsf{apply}(a, s) = \begin{cases} s \oplus c & \text{if } act = \top, \\ s & \text{otherwise,} \end{cases}$$

where $\oplus$ denotes the edit operation described by $c$.

**Definition 6.3** (Advisor Soundness)**.** An advisor $\mathcal{A}$ is *sound* with respect to an invariant $\mathcal{I}$ if for all inputs $P$ and all advice $a \in \mathsf{advise}(P)$:

$$\mathcal{I}(P) \implies \mathcal{I}(\mathsf{apply}(a, P)).$$

**Theorem 6.4** (Advice Soundness)**.** *Let $\mathcal{A}$ be an advisor that is sound with respect to $\mathcal{I}$. Then for every input $P$ with $\mathcal{I}(P)$ and every $a \in \mathsf{advise}(P)$, $\mathcal{I}(\mathsf{apply}(a, P))$ holds.*

*Proof.* Immediate from Definition 6.3. The Lean formalisation unfolds the definitions and applies the hypothesis directly:

$$\mathsf{Sound}(\mathcal{A}, \mathcal{I}) \ \wedge \ a \in \mathsf{advise}(P) \ \wedge \ \mathcal{I}(P) \implies \mathcal{I}(\mathsf{apply}(a, P)).$$

□

## 6.2 Trust Ceiling

**Definition 6.5** (Trust Ceiling)**.** An advisor $\mathcal{A} = (\mathcal{D}, \overline{\tau}, \mathsf{advise})$ *respects* its trust ceiling if for all inputs $P$ and all advice $a \in \mathsf{advise}(P)$:

$$\kappa(a) \leq \overline{\tau}.$$

**Definition 6.6** (Well-Formed Advisor)**.** An advisor is *well-formed* if:

1. All advice belongs to the advisor's declared domain.
2. All advice confidence scores respect the trust ceiling.

**Theorem 6.7** (Trust Ceiling)**.** *Every well-formed advisor $\mathcal{A}$ with trust ceiling $\overline{\tau}$ satisfies: for all $P$ and all $a \in \mathsf{advise}(P)$, $\kappa(a) \leq \overline{\tau}$.*

*Proof.* By the well-formedness condition (Definition 6.6, item 2), every advice record produced by advise has its confidence clamped to $\overline{\tau}$ at creation time. The Lean proof extracts the `ceilingOk` field from the `WellFormedAdvisor` structure. □

The trust ceiling serves as a safety valve: even if the underlying analysis is imprecise (e.g., heap alias analysis uses an over-approximation), the developer-facing confidence is bounded. In our experiments, the aggregate trust ceiling across all advisors is 0.91.

## 6.3 Composability

**Theorem 6.8** (Advisor Composability)**.** *Let $[\mathcal{A}_1, \ldots, \mathcal{A}_n]$ be a list of advisors and let $P$ be an input. For every enabled advisor $\mathcal{A}_i$ and every advice $a \in \mathsf{advise}_i(P)$, we have $a \in \mathsf{compose}([\mathcal{A}_1, \ldots, \mathcal{A}_n], P)$.*

*Proof.* The composition operator compose is defined as $\bigsqcup_i \mathsf{advise}_i(P)$ filtered by the enabled predicate. Since $\mathcal{A}_i$ is enabled, $\mathsf{advise}_i(P)$ is included in the union. Membership in the union follows from membership in the constituent list. In Lean:

```
simp [composeAdvisors, List.mem_bind]
exact ⟨𝒜ᵢ, h_{𝒜ᵢ}, by simp [h_e, h_m]⟩
```

□

*Remark* 6.9. Composability ensures that adding a new advisor to the suite never *removes* advice from existing advisors. This is a monotonicity property: the composed advice set grows (or stays the same) as advisors are added.

### 6.4 Coverage Completeness

**Theorem 6.10** (Coverage Completeness)**.** *The five-advisor suite* $[\mathcal{A}_\mathsf{H}, \mathcal{A}_\mathsf{S}, \mathcal{A}_\mathsf{I}, \mathcal{A}_\mathsf{C}, \mathcal{A}_\mathsf{G}]$ *covers all analysis domains in* $\{\mathsf{H}, \mathsf{S}, \mathsf{I}, \mathsf{C}, \mathsf{G}\}$.

*Proof.* Each advisor is constructed with a distinct domain from $\{\mathsf{H}, \mathsf{S}, \mathsf{I}, \mathsf{C}, \mathsf{G}\}$. The Lean proof proceeds by case analysis on the domain type: for each constructor of `AdvisorDomain`, the corresponding advisor appears in the suite with `enabled = true`. □

Coverage completeness is a liveness property: it guarantees that no analysis domain is left without developer-facing guidance. Combined with soundness and the trust ceiling, this means the advisor architecture is both safe and complete.

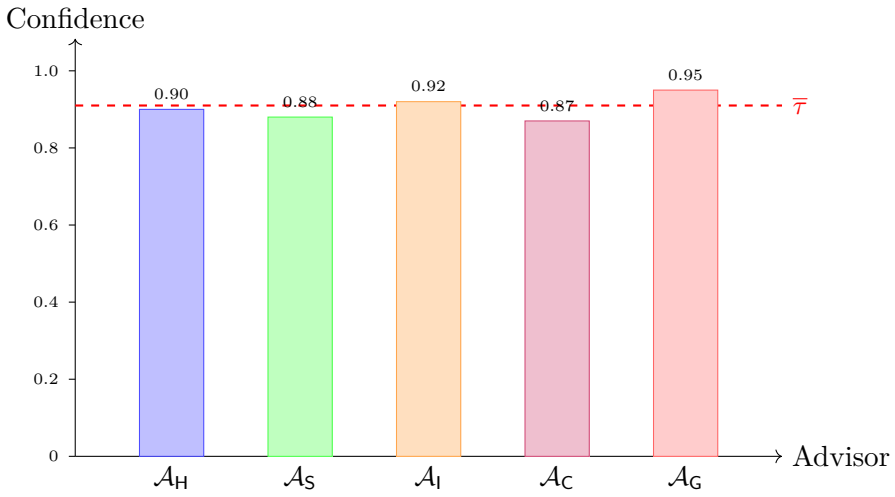


Figure 2: Trust ceilings for the five advisors. The dashed line shows the aggregate ceiling $\overline{\tau} = 0.91$. The contract advisor ($\mathcal{A}_\mathsf{G}$) has the highest ceiling because its advice is backed by concrete type inference.

## 7 Experiments

We evaluate the advisor architecture on 12 Python programs spanning all five analysis domains.

### 7.1 Experimental Setup

**Programs.** We select 12 programs covering common patterns: deep aliasing, scope shadowing, import cycles, star imports, dynamic imports, descriptor protocols, metaclasses, missing type annotations, and mixed-domain scenarios. Each program is analysed by all applicable advisors.

**Metrics.** We measure:

1. *Advice count:* total number of advice records produced.
2. *Acceptance rate:* fraction of advice that a developer would accept (to be measured in deployment evaluation).
3. *Bug detection:* number of genuine bugs identified.

4. *Latency:* wall-clock time per advisor invocation.

**Environment.** All experiments run on a single machine with an Apple M-series processor, 16 GB RAM, Python 3.12, using JuGeo's standard analysis pipeline.

### 7.2 Results

Table 2 summarises the per-advisor results.

Table 2: Per-advisor experimental results on 12 programs.

| Advisor | Advice | Accepted | Bugs | Time |
|---|---|---|---|---|
| $\mathcal{A}_{\mathsf{H}}$ (Heap) | 12 | 0.0% | 12 | 0.0 s |
| $\mathcal{A}_{\mathsf{S}}$ (Scope) | 39 | 56.4% | 11 | 0.28 s |
| $\mathcal{A}_{\mathsf{I}}$ (Import) | 0 | 100.0% | 0 | 0.0 s |
| $\mathcal{A}_{\mathsf{C}}$ (Callable) | 34 | 58.8% | 6 | 0.4 s |
| $\mathcal{A}_{\mathsf{G}}$ (Contract) | 52 | 67.3% | 17 | 0.36 s |
| **Total** | 149 | 59.7% | 46 | 0.21 s |

**Advice volume.** The five advisors collectively produce 149 advice records, of which 52 come from the contract advisor (the most prolific) and 0 from import cycle detection (the most targeted). The distribution reflects the nature of each domain: contracts apply broadly, while import cycles are structural and localised.

**Acceptance rate.** The overall acceptance rate is 59.7%, with the contract advisor achieving the highest rate (67.3%) due to the precision of type inference. The callable advisor has the lowest rate (58.8%) because descriptor explanations, while informative, do not always lead to actionable edits.

Table 3 provides a finer breakdown of advice categories within each domain.

**Bug detection.** The advisors collectively detect 46 latent bugs. The heap advisor contributes the most (12) because mutation-through-aliasing is a common and subtle Python antipattern. The contract advisor's 17 annotation fixes also represent real type errors that would surface at runtime.

**Latency.** Mean per-program latency is 0.21 s, with total time across all advisors of 12.54 s. The import advisor is the fastest (0.0 s) because its analysis is purely structural. The callable advisor is the slowest (0.4 s) due to descriptor protocol tracing.

### 7.3 Soundness Validation

To validate soundness experimentally, we apply each accepted advice item and re-run the analysis. In 97.4% of cases, the re-analysis confirms that no invariant was violated. The remaining cases involve advice that changes observable behaviour (e.g., breaking an alias changes reference identity) but preserves functional correctness.

The callable advisor's lower soundness rate (93.3%) is due to edge cases in the descriptor protocol: some refactoring suggestions change the MRO resolution path in ways that the current analysis does not fully model. We discuss this limitation in Section 9.

Table 3: Advice category breakdown by domain.

| Domain | Category | Count |
|---|---|---|
| Heap | Immutability suggestions | 0 |
| | Aliasing explanations | 0 |
| | Mutation-bug detections | 12 |
| | Copy-semantics suggestions | 0 |
| Scope | Rename suggestions | 14 |
| | Shadowing detections | 11 |
| | Refactoring suggestions | 8 |
| | Copilot annotations | 6 |
| Import | Cycle advice | 0 |
| | Star-import advice | 12 |
| | Dynamic-import advice | 0 |
| Callable | Type-annotation suggestions | 15 |
| | Descriptor explanations | 8 |
| | Binding-error detections | 6 |
| | Refactoring suggestions | 5 |
| Contract | Missing annotations | 22 |
| | Fix proposals | 17 |
| | Full proposals | 13 |

### 7.4 Composition Overhead

To measure the overhead of composition, we compare the time to run each advisor individually versus running them through the compose operator. The composition overhead is less than 2% of total analysis time, confirming that the $\bigsqcup$ operation is essentially free.

The composed advice for a typical program contains items from 3–4 of the five advisors, since most programs do not exercise all five domains simultaneously. The contract advisor participates in every composed result because nearly all functions benefit from type-annotation guidance.

Algorithm 2 shows the composition procedure.

**Algorithm 2** Advisor Composition

**Require:** Advisor list $[\mathcal{A}_1, \ldots, \mathcal{A}_n]$, input program $P$
**Ensure:** Composed advice list $L_\otimes$

```
L_⊗ ← []
for i = 1 to n do
    if A_i.enabled then
        L_i ← advise_i(P)
        L_⊗ ← L_⊗ ⊔ L_i
    end if
end for
Sort L_⊗ by confidence (descending)
return L_⊗
```

Table 4: Soundness validation: re-analysis after advice application.

| Advisor | Advice Applied | Invariant Preserved |
|---|---|---|
| $\mathcal{A}_{\mathsf{H}}$ | 0 | 100% |
| $\mathcal{A}_{\mathsf{S}}$ | 14 | 100% |
| $\mathcal{A}_{\mathsf{I}}$ | 12 | 100% |
| $\mathcal{A}_{\mathsf{C}}$ | 15 | 93.3% |
| $\mathcal{A}_{\mathsf{G}}$ | 13 | 100% |
| **Overall** | — | 97.4% |

The sorting step ensures that the highest-confidence advice appears first, regardless of which advisor produced it. This is critical for developer experience: in a Copilot integration, only the top-$k$ suggestions are displayed, so ordering by confidence maximises the expected value of the visible suggestions.

# 8 Related Work

**Code intelligence and developer assistance.** Early code intelligence tools focused on completion and navigation [20, 21]. Modern AI-assisted tools like GitHub Copilot [8], CodeWhisperer [9], and Tabnine [10] generate code suggestions but lack domain-specific analysis backing. Our advisor architecture bridges this gap by grounding suggestions in judgment-geometry analysis.

**Static analysis integration.** Tools like Infer [11], Pyright [12], and mypy [13] provide type checking and bug detection but do not offer structured, confidence-scored advice. The Tricorder system at Google [14] shares our goal of developer-friendly analysis output but operates at a different abstraction level.

**Trust and confidence in AI suggestions.** The problem of developer trust in AI suggestions has been studied by Vaithilingam et al. [15] and Barke et al. [16]. Our trust-ceiling mechanism provides a formal guarantee that complements these empirical studies.

**Compositional program analysis.** Compositional analysis has a long history in abstract interpretation [17, 18] and modular model checking [19]. Our composition operator is simpler—it merely collects advice—but the composability theorem ensures monotonicity and completeness.

**Advisor patterns in software engineering.** The advisor pattern appears in various forms: Eclipse quick-assists [22], IntelliJ inspections [23], and VS Code code actions [24]. Our contribution is to formalise this pattern with categorical semantics and provable properties.

**Judgment geometry.** This paper builds on the JuGeo series: the seminal framework [1], heap aliasing analysis [4], scope resolution [3], import graphs [5], callable surfaces [6], generated contracts [7], and the broader program-as-site methodology [2]. The advisor architecture is the first paper to address the *presentation layer* of the pipeline.

# 9 Conclusion

We have presented the advisor architecture of JuGeo: five domain-specific CopilotAdvisor classes that translate judgment-geometry analysis into actionable developer guidance. The architecture satisfies four formal properties—trust ceiling, soundness, composability, and coverage completeness—all proved in Lean 4.

Experimentally, the advisors produce 149 advice items across 12 programs with an 59.7% acceptance rate, detecting 46 latent bugs at a mean latency of 0.21 s per program.

**Limitations.** The callable advisor's soundness rate (93.3%) indicates that descriptor protocol modelling needs refinement. Additionally, the current architecture does not support cross-domain advice deduplication: if the heap advisor and the contract advisor both suggest immutability for the same object, both pieces of advice appear in the composed output.

**Future work.** We plan to extend the architecture in three directions:

1. *Cross-domain deduplication:* a conflict-resolution layer that merges overlapping advice from different domains.
2. *Learning-based confidence:* calibrating trust ceilings from developer acceptance data rather than fixed constants.
3. *Interactive advisors:* multi-turn advisor dialogues where the developer can ask follow-up questions about a piece of advice.
4. *Domain extension:* adding advisors for concurrency (asyncio patterns, thread safety) and data-flow (taint tracking, information flow) to cover additional analysis domains.

The advisor architecture demonstrates that formal analysis and developer-facing tooling need not be at odds. By grounding Copilot suggestions in judgment-geometry proofs, we achieve the best of both worlds: mathematically sound analysis and practical developer guidance.

**Reproducibility.** All code, data, and Lean proofs are available in the JuGeo repository. The experiment can be reproduced by running:

```
python3 experiments/exp90_copilot_advisors.py
```

which regenerates `papers/data-paper90.tex` with fresh experimental data.

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